

Prepared By	09-Jun-2012	Mr.Chatchawan S.	 RENEWCOILS ENGINEER & SUPPLY CO.,LTD. Renew coils Engineering & Supply Co., Ltd.	
Checked By	09-Jun-2012	Mr.Chatchawan S.		
Approved By	09-Jun-2012	Mr.Amnauy W.		
Certified By	09-Jun-2012	Mr.Amnauy W.		
Rev.	Date	Checked	Approved	Description
0	09-Jun-2012	Mr.Chatchawan S.	Mr.Amnauy W.	Issue for Approved

Renew Coils Engineering & Supply Co., Ltd.

Doc. No. : RNC-WPS-001-2012

Addendum

**WELDING PROCEDURE SPECIFICATION (WPS)
RNC-WPS-005**

WELDING PROCEDURE SPECIFICATION (WPS)

Company Name **Renew Coil Engineering & Supply Co., Ltd.**
WPS No. **RNC-WPS-005** Date **9-Jun-12** Revision No. **0**
Supporting PQR No.(s) **RNC-PQR-005** Date **9-Jun-12** Type(s) **Manual**
Welding Process(es) **Gas Tungsten Arc Welding (GTAW)** (Automatic, Manual, Machine, or Semi-Auto.)

JOINT (QW-402)

Joint Design **Single V**
Backing (Yes) ☐ (No) ☒ X
Backing Material (Type) **N/A**

☐ Metal ☐ Nonfusing Metal
☐ Nonmetallic ☐ Other

Notes _____

See Details Attachment sheet Joint Sketch

BASE METALS (QW-403)

P-No. **1** Group No. **1** to P-No. **1** Group No. **1**
Material Specification Type and Grade **SA 333 Gr.1,6 SA420 WPL 6** TO **SA 333 Gr.1,6 SA420 WPL 6**
Chem. Analysis and Mech. Prop. _____
Thickness Range :
Base Metal : Groove **1.6 mm - 7.82 mm.** Fillet **All**
Pipe Dia. Range : Groove **All** Fillet **All**
Other _____

FILLER METALS (QW-404)

	GTAW
Spec. No. (SFA)	5.28
AWS No. (Class)	ER-70S-G
F-No.	6
A-No.	1
Size of Filler Metals	Ø 2.0 & 2.4 mm
Deposited Weld Metals	3.91 mm
Thickness Range :	
Groove	7.82 mm Max.
Fillet	All
Electrode-Flux (Class)	-
Flux Trade Name	-
Consumable Insert	-
Other	KobeTGS-1N or Equivalents

POSITION (QW-405)

Position(s) of Groove **ALL**
Welding Progression Up **X** Down _____
Position(s) of Fillet **ALL**

POSTWELD HEAT TREATMENT (QW-407)

Temperature Range **593 - 649°C**
Time Range **Holding Time 1 Hour**
Notes **Heating & Cooling Rate = 200 ° Max.**

PREHEAT (QW-406)

Preheat Temp. Min. **10°C**
Interpass Temp. Max. **238°C**
Preheat Maintenance **N/A**
Notes _____

GAS (QW-408)

	Percent Composition		
	Gas(es)	(Mixture)	Flow Rate
Shielding	Argon	99.99%	13-17Ltr/Min
Trailing	-	-	-
Backing	-	-	-

WELDING PROCEDURE SPECIFICATION (WPS)

WPS No. RNC-WPS-005 Rev. 0

ELECTRICAL CHARACTERISTICS (QW-409)

Current ☐ AC ☒ DC Polarity EN
 Amps (Range) 80-120 Volts (Range) 10-13
 Tungsten Electrode Size and Type Ø 2.4 mm, Thoriated (EWTh-2)
 (Pure Tungsten, 2% Thoriated, etc.)
 Mode of Metal Transfer for GMAW N/A
 (Spray arc, short circuiting arc, etc.)
 Electrode Wire feed speed range N/A

TECHNIQUE (QW-410)

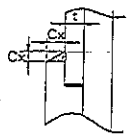
String or Weave Bead BOTH APPLICATIONS
 Orifice or Gas Cup Size 8.0 mm - 19.0 mm
 Initial and Interpass Cleaning (Brushing, Grinding, etc.) BRUSHING OR GRINDING
 Method of Back Gouging N/A
 Oscillation N/A
 Contact Tube to Work Distance N/A
 Multiple or Single Pass (per side) MULTIPASS
 Multiple or Single Electrodes SINGLE ELECTRODE
 Travel Speed (Range) 5.0-8.0Cm/Minute
 Peening N/A
 Other N/A

Weld Layer(s)	Process	Filler Metal		Current		Volt Range	Travel Speed Range (Cm/min)	Other
		Class	Dia. (mm)	Type Polar.	Amp. Range			
1st	GTAW	ER70S-G	2.4	DC EN	80-100	10-12	5.0-7.0	
2nd & Other(s)	GTAW	ER70S-G	2.4	DC EN	90-120	11-13	6.0-8.0	

We, the undersigned, certify that the statements in this record are correct and that the test welds were prepared, welded, and tested in accordance with the requirement of **ASME-IX, 2007 latest Revision**.

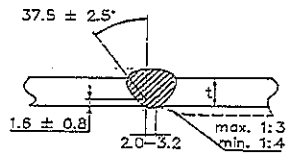
Prepared by: Mr. Chatchawan S. 9/06/12
 Name Signed Date
 Reviewed by: Mr. Saksan Ch. 9-06-12
 Name Signed Date
 Approved by: ANNUAY W. 9-06-12
 Name Signed Date

ATTACHED FIGURE JOINT SKETCH



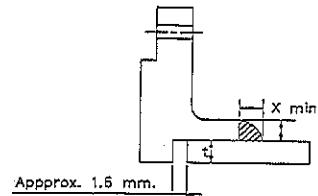
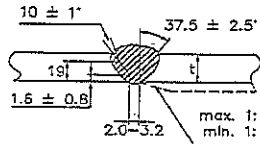
$Cx \text{ (min.)} = 1.25t \text{ but not less than } 3.2 \text{ mm.}$

JOINT No. A1



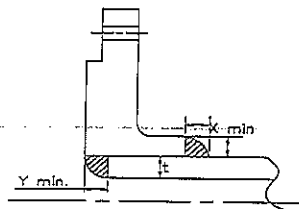
Misalignment (max.) = 1.5 mm.

JOINT No. B1 & B2



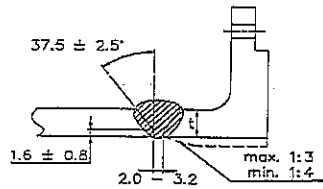
X min. = The lesser of $1.4t$ or thickness of the hub.

JOINT No. A2



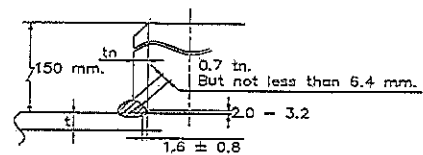
X min. = The lesser of $1.4t$ or thickness of the hub.
Y min. = The lesser of t or 6.4 mm.

JOINT No. A3



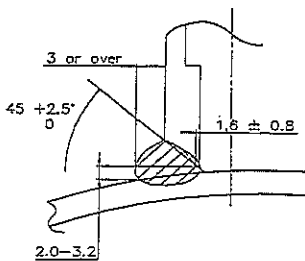
Misalignment (max.) = 1.5 mm.

JOINT No. B3

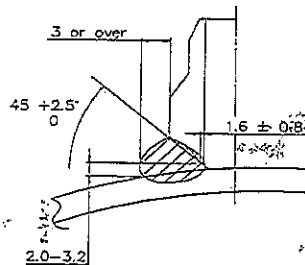


Misalignment (max.) = The lesser of 3.2 mm. or $0.5tn$.

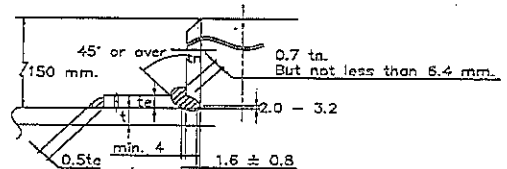
JOINT No. B4



JOINT No. C1

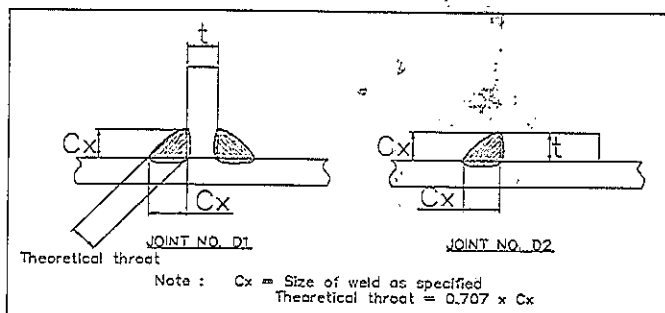


JOINT No. C2



Misalignment (max.) = The lesser of 3.2 mm. or $0.5tn$.

JOINT No. B5





Procedure Qualification Record (PQR)

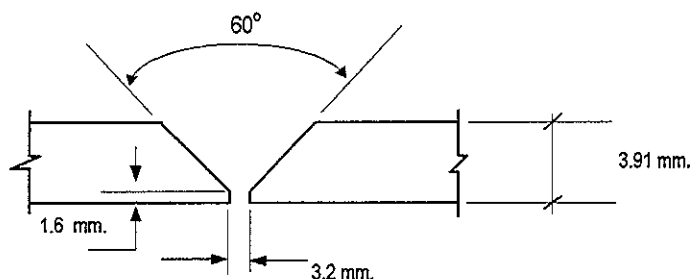
ASME-QW-200.2, Section IX, ASME Boiler Pressure Vessel Code

Record Actual Variables Used to Weld Test Coupon

RECORD OF WELDING (QW-483) PQR Number : PQR-RNC-005

Company Name :	RENEWCOILS ENGINEERING & SUPPLY CO.,LTD.		
PQR Record Number :	RNC-PQR-005	Date :	13 June 2012
WPS Number :	RNC-WPS-005		
Welding Process (es) :	Gas Tungsten Arc Welding (GTAW)		
Type (Manual, Semi-auto, or Automatic) :	Manual		

JOINT DESIGN (QW-402)



(For combination qualifications, the deposited weld metal thickness shall be recorded for each filler metal or process used)

WELDING VARIABLES

BASE METALS (QW-403)				POSTWELD HEAT TREATMENT (QW-407)			
Material Spec.	SA333	To	SA333	Temperature	610 °C		
Type or Grade	Gr.6	To	Gr.6	Hold Time	1 HR.		
P. No.	1	To	1	Others	GAS (QW-408)		
Thickness of Test Coupon	3.91 mm.						
Diameter of Test Coupon	Ø 2"						
Others							
FILLER METALS (QW-404)				Percent Composition			
					Gas (es)	Mixture	Flow Rate
				Shielding	Argon	99.95%	15 L/Min.
				Trailing	-	-	-
				Backing	-	-	-
ELECTRICAL CHARACTERISTICS (QW-409)				Current DC			
				Polarity EN			
				Amperage	81 - 94	Voltage	10 - 11
				Tungsten Electrode Size	ø 2.4 mm. 2% thoriated.EWTh-2		
				Others	N/A		
TECHNIQUE (QW-410)				Travel Speed 5 - 6 cm/min.			
				String or Weave Bead Both			
				Oscillation N/A			
				Multipass or single Pass Multiple Pass			
				Multipass or Single Electrodes Single Electrode			
PREHEAT (QW-406)				Others _____			
				N/A			
POSITION (QW-405)							
Position of Groove	6G						
Progression of Welding (Uphill or Down)	Up Hill						
Others	N/A						
PREHEAT (QW-406)							
Preheat Temperature	Ambient Temperature						
Interpass Temperature (max.)	219 °C						
Others	N/A						

Procedure Qualification Record

ASME-QW-200.2, Section IX (WPQR)

RECORD OF TEST RESULTS (QW-483) PQR Number:- RNC-PQR-005

Tensile Test Report No.TS-12-06-102

Specimen	Width (mm)	Thickness (mm)	Area (mm ²)	Ultimate Total Load (KN)	Ultimate Strength (N/mm ²)	Type of Failure & Loc.
2363/12 TR1	12.63	3.79	47.868	22.320	466.283	OUT OF WELD
2363/12 TR2	13.09	3.70	48.433	22.948	473.800	OUT OF WELD

Guided Bend Tests (QW-160) Report No.BD-12-06-100

Type and Figure Number	Type of Bend	Indication	Results
2363/12 BF1	Transverse face bend	No Open Discontinuity	Accepted
2363/12 BF2	Transverse face bend	No Open Discontinuity	Accepted
2363/12 BR1	Transverse root bend	No Open Discontinuity	Accepted
2363/12 BR2	Transverse root bend	No Open Discontinuity	Accepted

Toughness Tests (QW-170)

Specimen Number	Notch Location	Notch Type	Test Temp.	Impact Values	Lateral Expansion		Drop Weight	
					% Shear	Mils	Break	No Break
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Fillet Weld Test (QW-180)

Result - acceptable : Yes N/A No N/A Penetration Into Base Metal : Yes N/A No N/A
Macroetch Test Results N/A

Other Tests

Type of Test Radiographic examination results Accepted Report No.PQR -001

Hardness Testing Report No. HN-12-06-077

Macrostructure Report No. MA-12-06-099

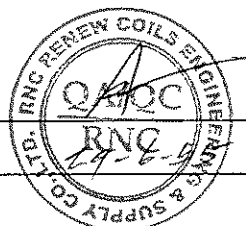
Impact Testing Report No.IM-12-06-136

Welders Name Mr.Niwat R. Welder Number -
Tests Conducted By Mr.Keerati Sa-ngoen Laboratory Test LPN PLATE MILL

We certify that the statements in this record are correct and that the test coupons were prepared, welded, and tested in accordance with the requirements of section IX of the ASME Boiler and Pressure Vessel Code.

Manufacture or Contractor :

Date :



Inspected by :

Certified by :

Date :

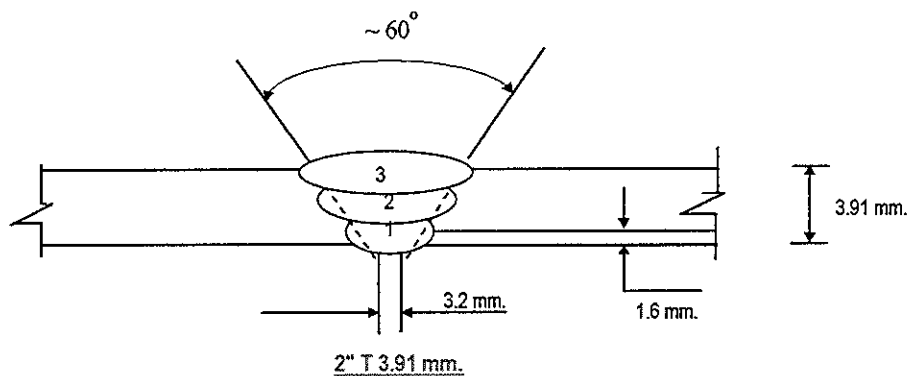
Mr.Nathawut M.

Mr.Chaloemkiet R.



PAE TECHNICAL SERVICE	WELDING PROCEDURE	PQR No. : RNC-PQR-005
		WPS No. : RNC-WPS-005
	QUALIFICATION RECORDS	THIS PQR RECORDS ACTUAL TEST CONDITION AND RESULT

Welding Sequences for PQR-RNC-005 (Butt Joint)



Weld Bead No.	Process	Class Filter Metal	Wire Size (mm)	Arc Current (Amps)	Arc Voltage (Volts)	Pot. AC/DC+	String or Weave Bead	Wire Feed Speed	Rate of Travel (cm/min)	Heat Input (KJ/cm)	Gas Shielding Flow Rate (l/min)	Gas Backing Flow Rate (l/min)
1	GTAW	ER70S-G	2.4	81	10	DCEN	-	-	5	9.72	15	-
2	GTAW	ER70S-G	2.4	91	11	DCEN	-	-	6	10.01	15	-
3	GTAW	ER70S-G	2.4	94	11	DCEN	-	-	6	10.34	15	-

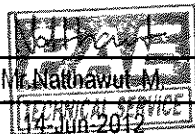
Page No. 1 of 1

PAE Inspection By

SIGNATURE :

NAME :

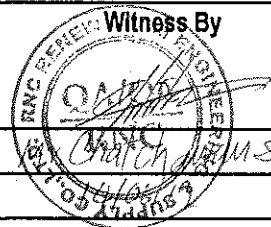
DATE :



SIGNATURE :

NAME :

DATE :



Witness By

SIGNATURE :

NAME :

DATE:

A

29-6-12



PAE TECHNICAL SERVICE CO.,LTD

NDT, TESTING, INSPECTION & ENGINEERING CONSULTANT

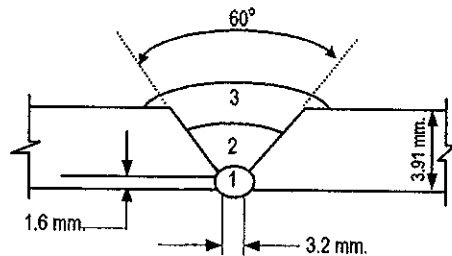
69 SOLOONNUT 64 (SUKSAMARN) SRINAKARIN RD, SUANLUANG, BANGKOK 10250
Tel : 02) 322-0222, Fax : 02) 721-2577, RAYONG (038) 607-402-3

WELDER PROCEDURE QUALIFICATION RECORD

Page No. 1 of 1

Client : Renew Coils Engineering & Supply Co.,Ltd.		Project Name : PQR		Location : RNC Work Shop		Report No. PQR-001 A.			
		Customer Order No. : N/A		WPS No. : RNC-WPS-005		Date : 28-Jun-12			
Welder No.	Welder Name	Welding Process	Test Position	Applicable Standard	Material		Electrode Specification	Results	
					Spec	Size		Visual	RT
-	MR.NIWAT R.	GTAW	6G	ASME IX	A333Gr.6	ø 2" Thk. 3.91 mm.	ER70S-G	ACCEPTED	ACCEPTED

Joint Details



Layer (S)	Process	Filler Metal		Current		Volt	Travel Speed cm/min	Interpas Temp °C	Heat Input (KJ / cm)	Shielding		Backing	
		Class	Dia. (mm.)	Type Polarity	Amp					GAS	Flow Rate (l/min)	GAS	Flow Rate (l/min)
1	GTAW	ER70S-G	2.4	DCEN	82	10	6	151	8.20	Argon	15	-	-
2	GTAW	ER70S-G	2.4	DCEN	92	11	7	185	8.62	Argon	15	-	-
3	GTAW	ER70S-G	2.4	DCEN	94	11	6	212	10.34	Argon	15	-	-

PAE Inspection By

Witness By

Approved By

SIGNATURE :

NAME :

DATE :

Mr. Nathawut M.
29-Jun-2012

SIGNATURE :

NAME :

DATE :

Mr. Chatchan M.S.
29-Jun-2012

SIGNATURE :

NAME :

DATE :

29-06-12



Renew Coils Engineering & Supply CO., LTD.

59 MOO 5 HIGHWAY 36, T. TABMA, A. MUANG, RAYONG 21000

TEL : (038) 663 - 476 FAX : (038) 633 - 477 MOBILE : (086) 340 - 9978

Radiographic Examination Report

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Renew Coils Engineering & Supply CO., LTD.

59 MOO 8 HIGHWAY 36, T. TABMA, A. MUANG, RAYONG 21000

TEL : (038) 663 - 476 FAX : (038) 633 - 477 MOBILE : (086) 340 - 9978

Mechanical Test Report



LPN PLATE MILL

Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjeree, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415 , E-mail : qa@lpnrm.co.th



21710

TESTING
No. 010

Tension Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Page 1 of 1

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Report No. TS-12-06-102

Revision No. 0

Test Product	PQR No. RNC-PQR-005	Material Characterization	Pipe
Project Name	N/A	Test Method	ASTM A370-10 (Prepared by ASME IX : 2010)
Project No.	N/A	Standard of Test	ASME Sec. IX : 2010
Material Specification	A333 Gr.6	Type of Test	Reduced Section Tension
Dimension	Ø 2" Sch 40 (3.91 mm Thk.)	Equipment's Capacity	200 TONS
Date Received	19 June 2012	Equipment / Serial No.	SHIMADZU UH200A / 32098926
Date Tested	21 June 2012	Ambient Temperature	23 ± 5 °C

Item No.	Test No.	Specimen Dimension					Yield		Ultimate		Elongation (%)	Remark
		Diameter (mm)	Thickness (mm)	Width (mm)	Area (mm ²)	Gauge Length (mm)	Load (kN)	Strength (N/mm ²)	Load (kN)	Strength (N/mm ²)		
1	2363/12 TR1	N/A	3.79	12.63	47.868	N/A	N/A	N/A	22.320	466.283	N/A	OUT OF WELD
2	2363/12 TR2	N/A	3.70	13.09	48.433	N/A	N/A	N/A	22.948	473.800	N/A	OUT OF WELD

The unit of N/mm² is equivalent to MPa

Additional details : WPS No. RNC-WPS-005

Welding Process : GTAW

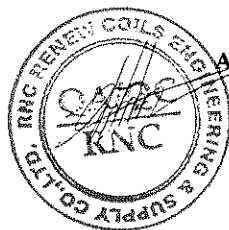
Filler Metal/Electrode Spec't : ER-70S-G/TGS-1N

Reported by :

(Mr.Keerati Sa-ngoen)

Laboratory Section Manager

Date 29 / 06 / 12



Approved by :

(Mr.Pavaret Preedawiphat)

Deputy Managing Director

Date 29 / 06 / 12

LPN PLATE MILL

Notes 1. The above results are valid exclusively for tested samples as mentioned in this report.

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LPN PLATE MILL

Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasatutjedee, Samutprakarn 10290 Thailand.

Tel 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnplm.co.th

21718

Bend Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Page 1 of 1

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Report No. BD-12-06-100

Revision No. 0

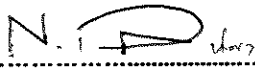
Test Product	PQR No. RNC-PQR-005	Material Characterization	Pipe
Project Name	N/A	Test Method	ASME Sec. IX : 2010
Project No.	N/A	Standard of Test	ASME Sec. IX : 2010
Material Specification	A333 Gr.6	Type of Jig	Guided-Bend Roller Jig
Dimension	Ø 2" Sch 40 (3.91 mm Thk.)	Diameter of Mandrel	16 mm
Date Received	19 June 2012	Bend Angle	180 Degree.
Date Tested	21 June 2012	Equipment / Serial No.	SHIMADZU UH200A/32098926

Item No.	Test No.	Type of Bend	Size of Specimen (ThickxWidthxLength)	Indication	Location of Indication	Remark
1	2363/12 BF1	Transverse face bend	3.95x19.06x159.96 mm	No Open Discontinuity	N/A	-
2	2363/12 BF2	Transverse face bend	3.92x19.06x160.59 mm	No Open Discontinuity	N/A	-
3	2363/12 BR1	Transverse root bend	3.98x19.03x159.31 mm	No Open Discontinuity	N/A	-
4	2363/12 BR2	Transverse root bend	3.99x19.09x159.87 mm	No Open Discontinuity	N/A	-

Additional details : WPS No. RNC-WPS-005

Welding Process : GTAW

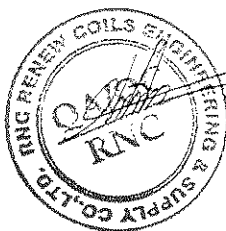
Filler Metal/Electrode Spec't : ER-70S-G/TGS-1N

Reported by : 

(Mr. Keerati Sa-ngoen)

Laboratory Section Manager

Date 29 / 06 / 12

Approved by : 

(Mr. Pavaret Preechawaphat)

Deputy Managing Director

Date 29 / 06 / 12

LPN PLATE MILL

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LPN PLATE MILL

Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand. 11062

Tel. 0 2815 6400-9 ext. 1179, 1210, Direct Line Fax. 0 2815 6415 , E-mail : qa@lpnpm.co.th

Macrostructure Analysis Report

Page 1 of 1

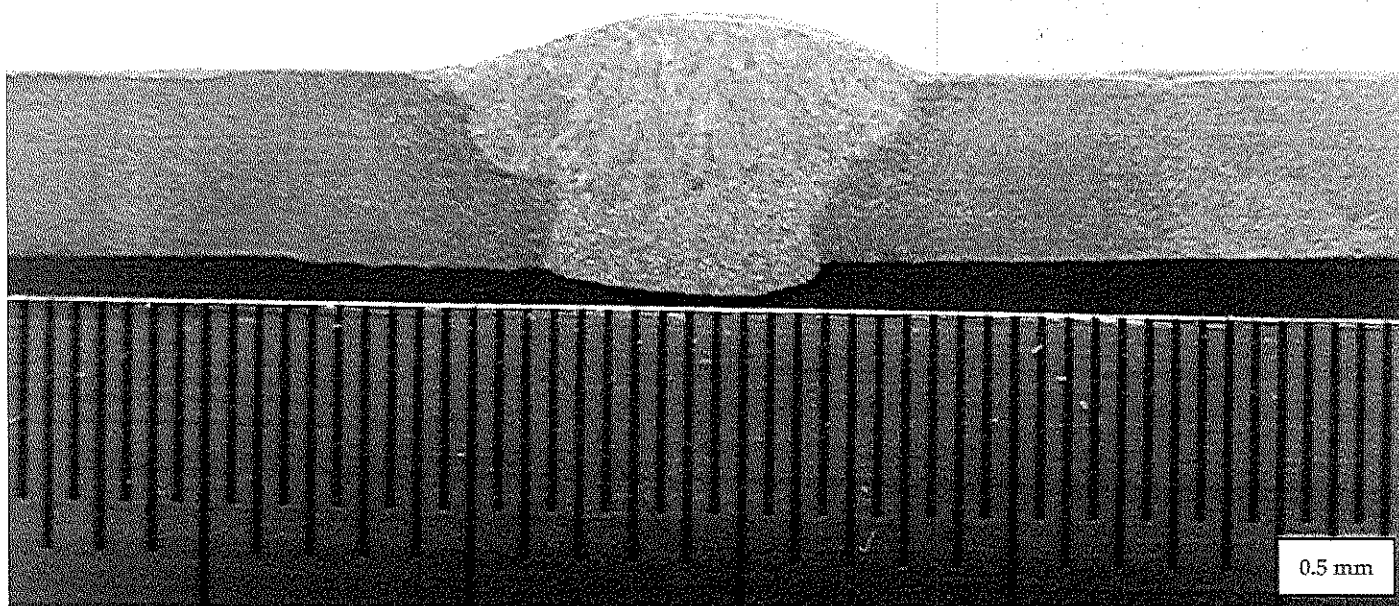
Customer : RENEWCOILS ENGINEERING & SUPPLY CO.,LTD.

Report No. MA-12-06-099

Address : 59 Moo 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000 Thailand.

Revision No. 0

Test Product	PQR No. RNC-PQR-005	Material Characterization	Pipe
Test No.	2363/12 MA	Test Method	ASME IX : 2010
Project Name	N/A	Standard of Test	ASME IX : 2010
Project No.	N/A	Etchant Reagent	75ml H ₂ O+25ml HNO ₃ (65%)
Material Specification	A333Gr.6	Temperature	28 °C
Dimension	Ø2"Sch40(3.91 mm Thk.)	Time of Etching	5 minutes
Welding Process/Position	GTAW	Surface Preparation	Polishing with sand paper
Type of Joint	Butt Joint		240, 400, 600, 800, 1000
Date Received	19 June 2012	Date Tested	21 June 2012

**Additional details :** The macrostructure shows the complete fusion. Weld metal and heat affected zone were free of cracks

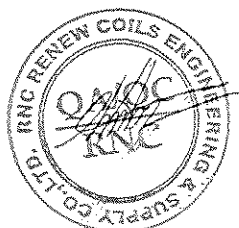
Filler/Electrode Spec't : ER-70S-G TGS-1N

Reported by :

(Mr. Keerati Sa-ngoen)

Laboratory Section Manager

Date : 29/06/12



Approved by :

(Mr. Pavaret Freedawiphat)

Deputy Managing Director

Date : 29/06/12

Notes

1. The above results are valid exclusively for tested samples as mentioned in this report.
2. Partial publicity of the results on testing is prohibited without the written permission from LPN LABORATORY.
3. This report shall not be reproduced in whole or in part without the written approval of LPN LABORATORY.

LPN PLATE MILL



LPN PLATE MILL

Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

12404

Tel. 0 2815 6400-9 ext. 1179, 1210, Direct Line Fax. 0 2815 6415 , E-mail : qa@lpnmpm.co.th

Hardness Testing Report

Page 1 of 1

Customer : RENEWCOILS ENGINEERING & SUPPLY CO.,LTD.

Report No. HN-12-06-077

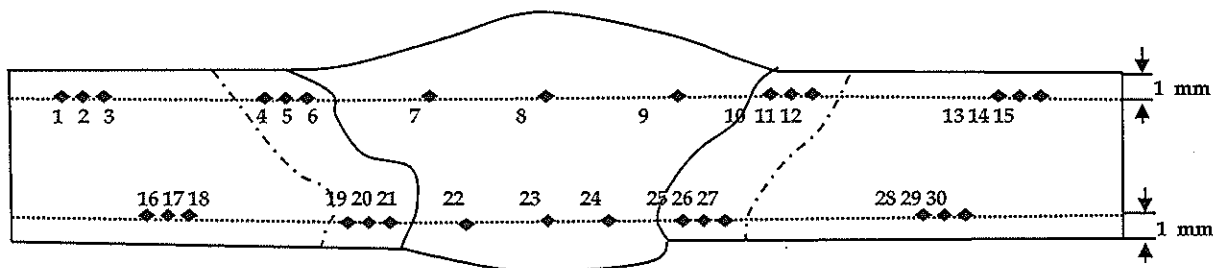
Address : 59 Moo 8 HIGHWAY 36 T.TABMA A.MUANG,RAYONG 21000 Thailand.

Revision No. 0

Test Product	PQR No. RNC-PQR-005	Material Characterization	Pipe
Test No.	2363/12 HN	Test Method	ASTM E92-82 : 2003
Project Name	N/A	Standard of Test	ASTM Sec. IX : 2010
Project No.	N/A	Type of test	Vicker Hardness Test
Material Specification	A333Gr.6	Equipment	Mitutoyo, HV-115
Dimension	Ø2"Sch40(3.91 mm Thk.)	Indenter Size	Diamond, pyramid 136 deg.
Welding Process/Position	GTAW	Test Load	HV 10 Kgf.
Type of Joint	Butt Joint	Total Force Dwell Time	15 s
Date Received	19 June 2012	Cal. Block	No. 62137 No. A62072
Date Tested	21 June 2012	Ambient Temperature	25.8 °C

Test Location

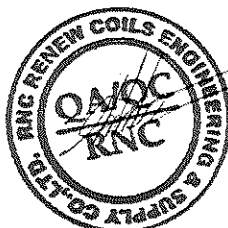
Total = 30 points



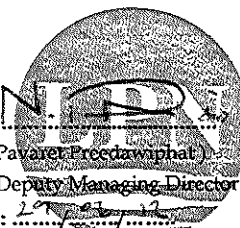
Point No.	Location	Hardness value Value (Scale HV)	Point No.	Location	Hardness value Value (Scale HV)	Point No.	Location	Hardness value Value (Scale)
1	BASE	133.9	16	BASE	130.1	*****		
2	BASE	132.2	17	BASE	130.8			
3	BASE	132.7	18	BASE	131.2			
4	HAZ	157.8	19	HAZ	160.6			
5	HAZ	171.3	20	HAZ	163.5			
6	HAZ	177.3	21	HAZ	168.8			
7	WELD	184.6	22	WELD	159.7			
8	WELD	181.8	23	WELD	162.0			
9	WELD	180.4	24	WELD	158.9			
10	HAZ	163.3	25	HAZ	165.4			
11	HAZ	153.3	26	HAZ	163.4			
12	HAZ	144.3	27	HAZ	155.6			
13	BASE	135.0	28	BASE	134.1			
14	BASE	134.3	29	BASE	132.7			
15	BASE	133.7	30	BASE	134.4			

Additional details : Filler/Electrode Spec't : ER-70S-G TGS-1N

Reported by : N. Sa-ngoen
 (Mr.Keerati Sa-ngoen)
 Laboratory Section Manager
 Date : 29/06/12



Approved by : N. Preechawiphat
 (Mr.Pavaret Preechawiphat)
 Deputy Managing Director
 Date : 29/06/12



Notes

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LPN PLATE MILL



LPN PLATE MILL

Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnrm.co.th



2864

Impact Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Page 1 of 1

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Report No. IM-12-06-136

Revision No. 0

Test Product	PQR No. RNC-PQR-005	Material Characterization	Pipe
Project Name	N/A	Test Method	ASTM A370-10 and ASTM E23M-07a ^{el}
Project No.	N/A	Standard of Test	ASME Sec. IX : 2010
Material Specification	A333 Gr.6	Type of Test	Charpy V-Notch Impact
Dimension	Ø 2" Sch 40 (3.91 mm Thk.)	Equipment / Serial No.	Tinius Olsen / 221811
Date Received	19 June 2012	Equipment's Capacity	542 Joules
Date Tested	22 June 2012	Ambient Temperature	23 ± 5 °C

Item No.	Test No.	Test Temperature (°C)	Size of Specimen			Results		Location of Test	Remark
			Thickness (mm)	Width (mm)	Length (mm)	Energy Absorbed (Joules)	Average (Joules)		
1	2363/12-1	-20°C	2.515	10.016	54.93	28.32	29.03	Base Metal	-
2	2363/12-2	-20°C	2.517	10.013	54.80	29.94			
3	2363/12-3	-20°C	2.528	10.018	54.79	28.83			
4	2363/12-1	-20°C	2.514	10.008	55.00	34.83	37.17	Weld Metal	-
5	2363/12-2	-20°C	2.514	10.005	54.95	39.07			
6	2363/12-3	-20°C	2.513	10.004	54.89	37.61			
7	2363/12-1	-20°C	2.508	10.008	54.76	43.35	44.58	HAZ	-
8	2363/12-2	-20°C	2.510	10.010	54.73	45.03			
9	2363/12-3	-20°C	2.508	10.011	54.60	45.35			

Additional details : WPS No. RNC-WPS-005

Welding Process : GTAW

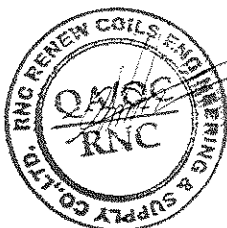
Filler Metal/Electrode Spec't : ER-70S-G/TGS-1N

Reported by :

(Mr.Keerati Sa-ngoen)

Laboratory Section Manager

Date30 / 07 / 12.....



Approved by :

(Mr.Pavaret Preedawiphat)

Deputy Managing Director

Date30 / 07 / 12.....

LPN PLATE MILL

Notes 1. The above results are valid exclusively for tested samples as mentioned in this report.

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Renew Coils Engineering & Supply CO., LTD.

59 MOO 8 HIGHWAY 36, T. TABMA, A. MUANG, RAYONG 21000

TEL : (038) 663 - 476 FAX : (038) 633 - 477 MOBILE : (086) 340 - 9978

Post weld Heat Treatment Report

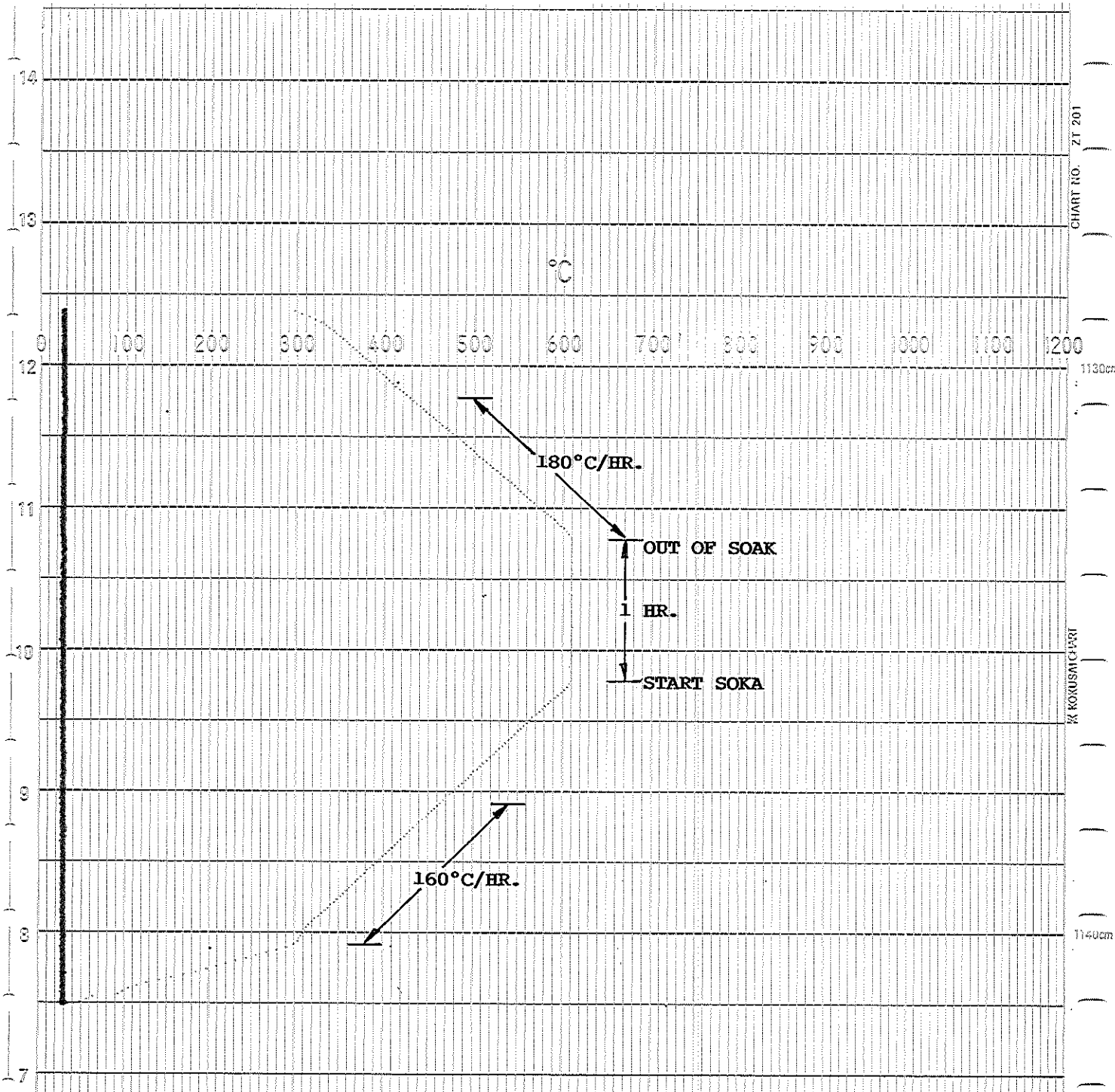
HEAT		J.S.T. HEAT TREATMENT COMPANY LIMITED	
POST WELD HEAT TREATMENT REPORT			
CLIENT : RENEW COILS ENGINEERING & SUPPLY CO.,LTD.			
CONTRACTOR :	J.S.T. HEAT TREATMENT CO.,LTD.		PAGE : 1
PROJECT : PQR			
P.W.H.T.METHOD : ☒ LOCAL ☐ FURNACE			
COMPLETION DATE : 02-07-2012			REPORT NO.PW -002
CHART SPEED : ☒ 25 MM/Hr, ☐ 50 MM/Hr, ☐ 100 MM/Hr.			
RECORDER NO : EH-066A041		MATERIAL : A333 GR.1,6 SA420 WPL6	
LOCATION : ☒ SHOP JST. ☐ FIELD		UNIT :	

HEAT TREATMENT CONDITION/RESULTS						
DESCRIPTION	LOADING TEMP(°C)	HEATING RATE (°C/Hr)	HOLDING TEMP (MIN-MAX)	HOLDING TIME (Hr)	COOLING RATE (°C/Hr)	UNLOADING TEMP (°C)
REQUEST	300	200	593-649	1 HR.	200	300
ACTUAL	300	160	610	1 HR.	180	300

DWG NO.,LINE NO.&CLASS,SHT.NO.SPOOL	WELD NO.	SIZE (")	THK. (MM)	WELD TYPE	T/C NO.
RNC-PQR-005		2"	3.91		3

TOTAL	1 EA.	JOINTS	TOTAL	D.B.
-------	-------	-------------------	-------	------

CHECKED BY J.S.T. HEAT TREATMENT CO.,LTD.		REVIEWED BY	APPROVED BY	APPROVED BY
SIGN :				
NAME :	SUPACHAI K.			
DATE :	02-07-2012			



DATE. 02-07-2012

RECORDER NO. EH - 066A041

CHART NO. PW - 002

CHART SPEED: 25mm./HR.

CUSTOMER. RENEW COILS ENGINEERING & SUPPLY CO., LTD.

PROJECT. PQR

DWG NO. RNC-PQR-005

SIZE 2" THK. 3.91mm

MATERIAL. A333 GR.1,6 SA420 WPL6

START TIME. 17.30 PM.

TECHNICIAN. MR. JIRAWAT S.

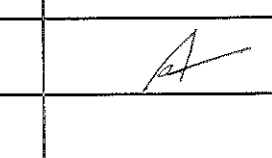
TC 3. COLOUR SKY BLUE



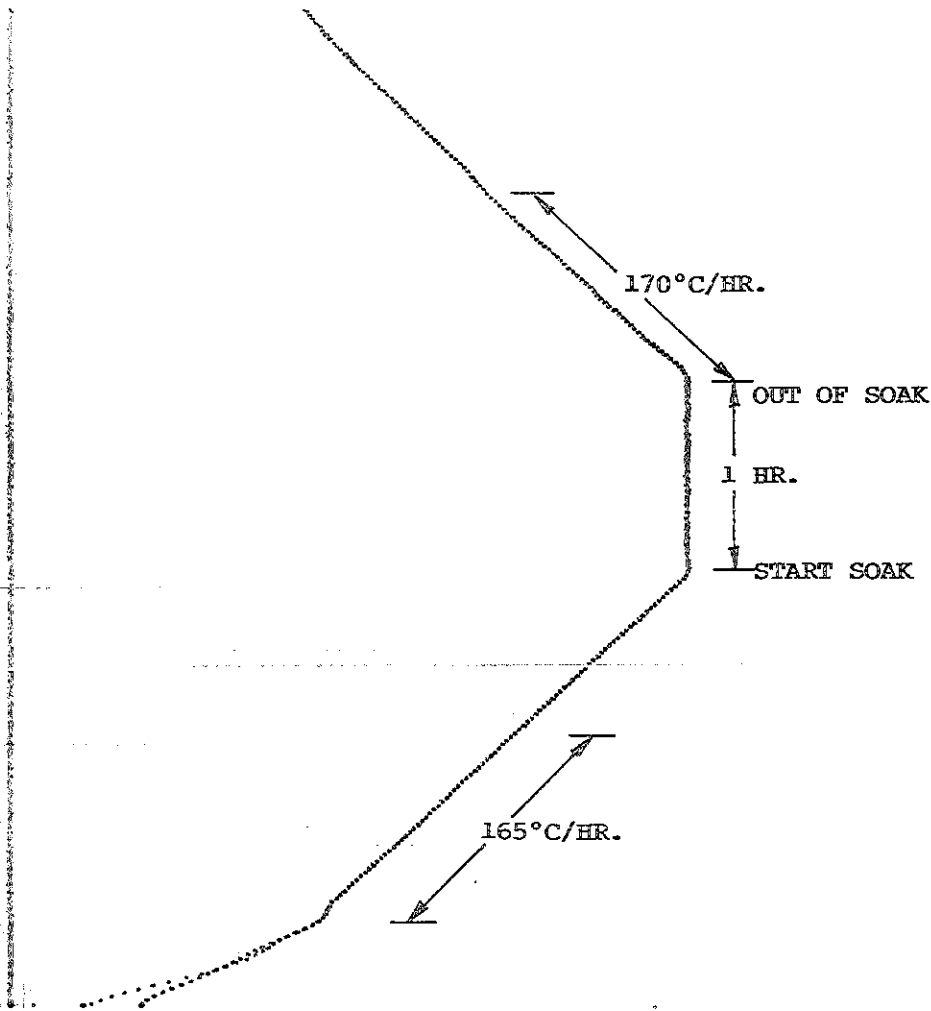
HEAT		J.S.T. HEAT TREATMENT COMPANY LIMITED	
POST WELD HEAT TREATMENT REPORT			
CLIENT : RENEW COILS ENGINEERING & SUPPLY CO., LTD.			
CONTRACTOR :	J.S.T. HEAT TREATMENT CO., LTD.		PAGE : 1
PROJECT : PQR			
P.W.H.T.METHOD : ☒ LOCAL ☐ FURNACE			
COMPLETION DATE : 14-06-2012			REPORT NO. PW-001
CHART SPEED : ☒ 25 MM/Hr, ☐ 50 MM/Hr, ☐ 100 MM/Hr.			
RECORDER NO : EH-066A042		MATERIAL : A335 GR.6, A106 GR.B	
LOCATION : ☒ SHOP JST. ☐ FIELD		UNIT :	

HEAT TREATMENT CONDITION/RESULTS						
DESCRIPTION	LOADING TEMP(C°)	HEATING RATE (C°/Hr)	HOLDING TEMP (MIN-MAX)	HOLDING TIME (Hr)	COOLING RATE (C°/Hr)	UNLOADING TEMP (C°)
REQUEST	300	200	593-649	1 HR.	200	300
ACTUAL	300	165	610	1 HR.	170	300

DWG NO., LINE NO. & CLASS, SHT. NO. POOL	WELD NO.	SIZE (")	THK. (MM)	WELD TYPE	T/C NO.
RNC-PQR-005		2"	3.91	BW	5
RNC-PQR-006		6"	7.11	BW	3,4
RNC-PQR-009		6"	10.97	BW	1,2
TOTAL	3	JOINTS	TOTAL	14	D.B.

CHECKED BY J.S.T. HEAT TREATMENT CO., LTD.		REVIEWED BY	APPROVED BY	APPROVED BY
SIGN :				
NAME :	S. PACHAI K.	M. CHANDRA S.		
DATE :	14-06-2012		29-06-12	

280cm



270cm

DATE. 14-06-2012

RECORDER NO. EH - 066A042

CHART NO. PW - 001

CHART SPEED. 25mm./HR.

CUSTOMER. RENEW COILS ENGINEERING & SUPPLY CO., LTD.

PROJECT. PQR

DWG NO. RNC-PQR-005

SIZE. 2" THK. 3.91mm.

DWG NO. RNC-PQR-006

SIZE. 6" THK. 7.11mm.

DWG NO. RNC-PQR-009

SIZE. 6" THK. 10.97mm.

MATERIAL. A333 GR.6, A106 GR.B

START TIME. 21.30 PM

TECHNICIAN. MR. ANU S.

TC	COLOUR
1.	RED
2.	BLACK
3.	SKY BLUE
4.	GREEN
5.	BROWN

280cm



COOPERHEAT

Certificate of Calibration / Test



INNOVATORS IN THERMAL TECHNOLOGY

Certificate No.

JST-053/2012

Date of Issue

1/2/2012

Signatory (Print)

Supachai K

Equipment Description

RECORDER "CHINO" EH SERIES

Input or Sensor Type

"K"

Output or Range

0-1200°C

Cooperheat Identification

Serial No.

Next Calibration Due

EH-066A042

1/8/2012

Customer Name

Customer Order No.

Works Order No.

Test Parameter	200	400	600	800	1000	1200
Prior to Test	200	400	600	800	1000	1200
Post Test	200	400	600	800	1000	1200

Instrument Used	Model	Model	Date Calibrated
FLUKE	714	6986010	25 June 2011

All measurement are performed with equipment that is traceable to National Standard unless otherwise

Signature

WEAT

Date

1/2/2012

Customer

J.S.T. HEAT TREATMENT CO., LTD.

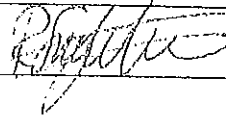
Certificate of Conformance

This material has been tested, inspected and approved by Cooperheat quality control in compliance with Cooperheat's BS EN ISO 9001 : 2000 Quality Management System.

43000 (100 mtrs)	TYPE "K" (NiCr-NiAl) Twin Thermocouple wire conforms to:- BS EN 60584-1: BS EN 60584-2: ANSI-MC96-1: Temperature Measurement Thermocouples. 1982
------------------	---

Batch No T11C7N1812/1142

QC Inspector's Sig:



Date

15/4/2008



CERTIFICATE OF CALIBRATION

mmc
(THAILAND) LTD

Issued By Marske Machine Hire & Calibration Centre

Date of Issue 25 June 2011

Certificate Number
E10544

Page 1 of 3 Pages



CALIBRATION
No. 0035

Marske Machine (Thailand) Co., Ltd
2nd Floor, The Seaboard Group Building
2/58 Highway 3191, MapTaPhut, Muang, Rayong 21150
Tel : (038) 682992-3, (038) 691563-4 Fax : (038) 682988
Email : sales@mmc-thai.com

Approved Signatory

[Signature]
Marut L

Customer : J.S.T. Heat Treatment Co., Ltd.
29/9 Soi Charoenmitr (Ekamai 10), Sukhumvit 63 Rd., Klongton Nue
Wattana, Bangkok 10110

Date Received : 24 June 2011

Instrument -	System ID :	EC00745	Job Number : 0000113262
	Description :	Thermocouple Calibrator	
	Manufacturer :	FLUKE	
	Model Number :	714	
	Serial Number :	6986010	Last Certificate Number : E00544
	Procedure Version :	EP0008	Last Calibration Date : 15/06/2010

Environmental Conditions

Temperature : 23°C +/- 2°C
Relative Humidity : 55% +/- 15%

Mains Voltage : 220V +/- 10V
Mains Frequency : 50Hz +/- 1Hz

Comments

The instrument was placed in lab and allowed to stabilise before calibrate.
The direct comparison technique has been applied to calibrate.
All tests passed Mfg'Spec Model 714 Rev 1,10/97.

Traceability Information

Instrument description	Serial number	Certificate number	Cal. Date	Cal. Period
3041 Precision Multi-Product Calibrator	F1031B8	19721	26/04/2011	52
3458A Multimeter	MY45043338	EE-0052-11	27/05/2011	52

Calibrated By : Tassanee K.

[Signature]

Date of Calibration : 25 June 2011

This certificate provides traceability of measurement to recognised National Standards, and to the units of measurement realised at the National Physical Laboratory or other recognised National Standards laboratories.
Copyright of this certificate is owned by the issuing laboratory and may not be reproduced except with the prior written approval of the issuing laboratory.
This certificate complies with the requirements of BS EN ISO 10012:2003

Quality Isn't Expensive It's Priceless

CERTIFICATE OF CALIBRATION



Certificate Number
E10544

Page 2 of 3 Pages

Test Title	Tolerance	STD Value	UUC Reading	% Of Spec.	Uncertainties
------------	-----------	-----------	-------------	------------	---------------

Calibration Information.

The uncertainty limits quoted refer to the measured values only, with no account being taken of the instrument ability to maintain its calibration.

The reported expanded uncertainty of measurement is state as the standard uncertainty of measurement multiplied by a coverage factor $k=2$ corresponding to a coverage probability of approximately 95%.

Function : DC Voltage Measurement

Range 75 mVDC	19.8uV	10.000 0mV	10.000mV	0%	4uV
Range 75 mVDC	28.8uV	-10.0000mV	-9.990mV	35%	6uV
Range 75 mVDC	19.8uV	30.000 0mV	30.010mV	51%	4uV
Range 75 mVDC	19.8uV	50.000 0mV	50.010mV	51%	4uV
Range 75 mVDC	19.8uV	70.000 0mV	70.010mV	51%	4uV

Function : Temperature Measurement

1370 °C T/C Type K	0.83°C	-100.00°C	-99.7°C	36%	0.17°C
1370 °C T/C Type K	0.76°C	0.00°C	0.3°C	39%	0.16°C
1370 °C T/C Type K	0.74°C	100.00°C	100.2°C	27%	0.15°C
1370 °C T/C Type K	0.73°C	500.00°C	500.2°C	27%	0.15°C
1370 °C T/C Type K	0.76°C	1 000.00°C	1 000.3°C	39%	0.16°C
1200 °C T/C Type J	0.74°C	-100.00°C	-99.6°C	54%	0.15°C
1200 °C T/C Type J	0.70°C	0.00°C	0.3°C	43%	0.14°C
1200 °C T/C Type J	0.68°C	100.00°C	100.3°C	44%	0.14°C
1200 °C T/C Type J	0.68°C	500.00°C	500.3°C	44%	0.14°C
1200 °C T/C Type J	0.67°C	1 000.00°C	1 000.3°C	45%	0.14°C
1760 °C T/C Type R	2.5°C	0.00°C	1.0°C	40%	0.50°C
1760 °C T/C Type R	1.8°C	100.00°C	100.0°C	0%	0.36°C
1760 °C T/C Type R	1.4°C	500.00°C	500.0°C	0%	0.28°C
1760 °C T/C Type R	1.3°C	900.00°C	900.0°C	0%	0.26°C
1760 °C T/C Type R	1.2°C	1 300.00°C	1 300.0°C	0%	0.24°C

Function : DC Voltage Source

Range 75 mVDC	28.8uV	9.998mV	10.00mV	8%	6uV
Range 75 mVDC	28.8uV	29.998mV	30.00mV	8%	6uV
Range 75 mVDC	28.8uV	49.997mV	50.00mV	10%	6uV
Range 75 mVDC	28.8uV	69.997mV	70.00mV	10%	6uV

Function : Temperature Source

1370 °C T/C Type K	0.83°C	-100.27°C	-100.0°C	33%	0.17°C
1370 °C T/C Type K	0.76°C	-0.24°C	0.0°C	32%	0.16°C
1370 °C T/C Type K	0.74°C	99.86°C	100.0°C	19%	0.15°C
1370 °C T/C Type K	0.73°C	499.85°C	500.0°C	21%	0.15°C
1370 °C T/C Type K	0.76°C	999.88°C	1 000.0°C	16%	0.16°C
1200 °C T/C Type J	0.74°C	-99.98°C	-100.0°C	3%	0.15°C
1200 °C T/C Type J	0.70°C	0.07°C	0.0°C	10%	0.14°C
1200 °C T/C Type J	0.68°C	100.04°C	100.0°C	6%	0.14°C
1200 °C T/C Type J	0.68°C	500.04°C	500.0°C	6%	0.14°C
1200 °C T/C Type J	0.67°C	999.95°C	1 000.0°C	7%	0.14°C

CERTIFICATE OF CALIBRATION



Certificate Number
E10544

Page 3 of 3 Pages

Item Title	Tolerance	STD Value	UUC Reading	% Of Spec.	Uncertainty
0 °C T/C Type R	2.5°C	0.75°C	0.0°C	30%	0.50°C
0 °C T/C Type R	1.8°C	99.50°C	100.0°C	28%	0.36°C
0 °C T/C Type R	1.4°C	499.54°C	500.0°C	33%	0.28°C
0 °C T/C Type R	1.3°C	899.75°C	900.0°C	19%	0.26°C
0 °C T/C Type R	1.2°C	1 299.84°C	1 300.0°C	13%	0.24°C

of calibration.



Renew Coils Engineering & Supply CO., LTD.

59 MOO 8 HIGHWAY 36, T. TABMA, A. MUANG, RAYONG 21000

TEL : (038) 663 - 476 FAX : (038) 633 - 477 MOBILE : (086) 340 - 9978

Consumable Certificate

TKW110801 KOBELCO

INSPECTION CERTIFICATE
(検 査 証 明 書)PURCHASER
需 要 家WELDING ROD OR WIRE
溶加棒又はワイヤCERTIFICATE No
証明書番号: 520113299
DATE OF ISSUE
発行日: 2011.10.28

TRADE DESIGNATION 品 名 (型 番)	DIMENSION 寸 法 (mm)	MFG.No 製造番号	APPLICABLE SPECIFICATION AND CLASSIFICATION 適用規格及び種類
TG-S1N	(2.4) mm 3/32 inch	GZ00820	AWS A5.28 ER70S-G ASME SFA-5.28 ER70S-G

CHEMICAL COMPOSITION 化学成分(%) ACCORDING TO EN10204 TYPE 3.1

ELEMENTS 成分	C	SI	MN	P	S	CU	NI	MO					
ROD OR WIRE 棒又はワイヤ	0.06	0.31	1.04	0.006	0.009	0.11	0.80	0.14					
ELEMENTS 成分									FS = FERRITE % (SCHAEFFLER DIAGRAM)				
ROD OR WIRE 棒又はワイヤ									FN = FERRITE NUMBER (DELONG DIAGRAM)				
									FNN = FERRITE NUMBER (WRC)				

TENSILE PROPERTIES OF DEPOSITED METAL 溶着金属の引張性能				IMPACT PROPERTIES OF DEPOSITED METAL 溶着金属の衝撃性能			HARDNESS PROPERTIES 硬さ
YIELD POINT 降伏点	YIELD STRENGTH AT 0.2% OFFSET 0.2% 耐力	TENSILE STRENGTH 引張強さ	ELONGATION 伸び	TEST TEMP. 試験温度	ABSORBED ENERGY 吸収エネルギー		
(-) N/mm ² ksi	(≥360) N/mm ² ≥52 ksi	(≥480) N/mm ² ≥70 ksi	≥24 %	(-60) °C -76 °F	AVG. 平均 (≥27) J	EACH (≥21) ≥20 ft-lbf ≥15	

WELDING CONDITIONS 溶接条件				POSTWELD HEAT TREATMENT 溶接後熱処理	
TYPE OF CURRENT 電流の種類	DOEN	SHIELDING GAS シールドガス	AR 100%		
AMPERAGE 溶接電流	220 A			(-) °C × - h	
ARC VOLTAGE アーク電圧	14 V			- °F × - h	

WE HEREBY CERTIFY THAT THE ABOVE TEST RESULTS
AND MECHANICAL PROPERTIES SPECIFICATIONS SATISFY
THE REQUIREMENTS OF THE APPLICABLE SPECIFICATION.

上記試験結果および機械的性質保証値は、適用規格の要求を
満足していることを証明します。

◆KOBELCO, LTD.
WELDING BUSINESS FUKUCHIYAMA PLANT
株式会社 神戸製鋼所 溶接事業部門 福知山工場

APPROVED BY CHIEF INSPECTOR
主任検査員 承認済

1110281407190728191110214104R2191

(注) 単位、適用規格・試験を除き大文字を使用しております。

REMARKS: CAPITAL LETTERS ARE USED EXCEPT FOR UNIT AND APPLICABLE SPECIFICATION / CLASSIFICATION.



Renew Coils Engineering & Supply CO., LTD.

59 MOO 8 HIGHWAY 36, T. TABMA, A. MUANG, RAYONG 21000

TEL : (038) 663 - 476 FAX : (038) 633 - 477 MOBILE : (086) 340 - 9978

Material Certificate

INSPECTION CERTIFICATE EN10204-3.1

Issued by
BENTELER STAHL/ROHR GMBH
RESIDENZSTR. 1
D-33104 PADERBORN,
GERMANY

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BENTELER

Stahl/Rohr

ABNAHMEPRÜFZEUGNIS EN 10204-3.1

INSPECTION CERTIFICATE EN 10204-3.1
CERTIFICAT DE RECEPTION EN 10204-3.1
EN 10204:2005-01

ROJPAIBOON EQUIPMENT CO.,LTD.
303,305,307,309 SIRINTORN RD.,
BANGBUMRHU, BANGPLAD,
BANGKOK 10700, THAILAND

Dokument-Nr.: 65-525631/001/P

Document No.:

No. du document:

Kunden-Bestell-Nr.: RJ811071RM

Purchase Order No.:

No. de commande du client:

Order confirmation no. 1538048

Versandanzeigen-Nr.: 6528487

Dispatch Note No.:

No. d'avis d'expédition:

Seamless carbon steel pipe ASTM A333 grade 6 / ASME
SA333 NACE MR0175 details as per order confirmation
no 1538048 date 12.11.2008

Prüf-Nr.:

Inspection No.:

No. du certificat:

Hersteller:

Manufacturer: BENTELER STAHL/ROHR GMBH

Producteur:

Herstellerzeichen:

Manufacturer's brand:

Marque du producteur:

Stempel des Abnahmebeauftragten: WA

Stamp of the inspection representative:

Poinçon du contrôleur:

Stahlschmelzungsverfahren: ELEKTROSTAHL
Steelmaking process: ELECTRIC FURNACE
Procédé d'élaboration de l'acier: FOUR ELECTRIQUE

L/C NUMBER: ML09000381

Lieferbedingungen:

Terms of delivery:

Conditions de livraison:

ASTM-A 333-2005-HF, ASME SA-333-HF, ASME Section II Part A Edition 2007 Addenda 2008, NACE
MR0175/ISO 15156-1:2001, NACE MR0175/ISO 15156-2:2003

Maße - Toleranzen:

Dimensions-tolerances:

Dimensions-tolérances:

length acc. to customer request, ASME B36.10M-2004, ASME SA-999, ASME Section II Part A Edition 2007 Addenda
2008, ASTM-A 999-2004a

Stahlsorte:

Steel grade:

Nuance d'acier:

GRADE 6

Lieferzustand:

Delivery condition:

État de livraison:

Normalized

Produktkennzeichnung:

Product marking:

Marquage du produit:

FS: BENTELER SIGN BENTELER A333-HF-SA333 6 SMLS SCHEDULE LT 50 WA TUBES-LENGTH METERS
DIMENSIONS HEAT-NO. NACE MR 0175/ISO 15156-1/2 - RM0811

AEZ = Ätzzeichenbeschriftung, Etching ink marking, Gravure à l'encre FK = Farbkennzeichnung, colour marking, marquage par couleur FS = Farbschablonierung, paint stencilling, marquage par peinture FSD = Farbstrahldrucker, Colour jet printer, imprimante à jet d'encre de c
LS = Längsschablonierung, longitudinal stencilling, marquage longitudinal par peinture PKE = Etikettenkennzeichnung, tag marking, marquage sur étiquette PS = Prägestempel, die stamp, marquage par poinçonnage TS = Tintenstrahlkennzeichnung, ink jet spray marking,

Pos.	Stück	Maße	Länge	Gewicht	Schmelzen-Nr.	Prüfdruck	Rohr-Nr.-Gruppe	Vielfachlängen
Item	Number	Dimensions	Length	Weight	Heat No.	Test pressure	Tube number group	Multiple lengths
Poste	Nombre	Dimensions	Longueur	Poids	No. de coulée	Pression d'épreuve	Série de no. des tubes	Longueurs multiples
		mm	m	kg				
0010	82	2" NPS * Sched. 40	6000	492,00	2664	552918		

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Pos.	Stück	Maße	Länge	Gewicht	Schmelzen-Nr.	Prüfdruck	Rohr-Nr.-Gruppe	Vielfachlängen
Item	Number	Dimensions	Length	Weight	Heat No.	Test pressure	Tube number group	Multiple lengths
Poste	Nombre	Dimensions	Longueur	Poids	No. de coulée	Pression d'épreuve	Série de no. des tubes	Longueurs multiples
		mm	m	kg				
0020	103	3" NPS * Sched. 40	6000	618,00	7067	762106		
0030	134	4" NPS * Sched. 40	6000	804,00	13070	762135		

Schmelzenanalyse [%] / Heat analysis [%] / Analyse sur coulée [%]

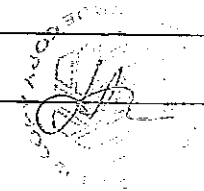
Pos.	Schmelzen-Nr.	C	SI	MN	P	S	AL
Item	Heat No.						
Poste	No. de coulée						
0010	552918	0,130	0,210	0,83	0,010	0,001	0,032
0020	762106	0,130	0,215	0,78	0,011	0,002	0,032
0030	762135	0,125	0,220	0,77	0,009	0,003	0,030


 GLOBAL CHESSON
 Chayut S.
 Inspection QA/QC Division
 Integrity and Reliability

Prüfresultate / Test results / Résultats des essais

Die Rohre wurden zerstörungsfrei geprüft: The tubes are non destructive tested: Les tubes ont passé un essai non destructif:	ET-test: acc. to ASME SA-999, ASTM-A 999; acc. to ASTM-E 309; imperfections	PASSED
Augensichtkontrolle: Visual inspection: Examen visuel:	PASSED	PASSED
Maßkontrolle: Dimensions examination: Vérification des dimensions:	PASSED	PASSED
Ringfaltversuch: Flattening test: Essai d'aplatissement:	PASSED	PASSED

Ergebnisse der mechanischen Prüfung / Results of mechanical testing / Résultats des essais mécaniques

 Die Probennahme erfolgte an Vielfachlängen.
 The sampling was carried out on multiple lengths.
 L'échantillonnage était réalisé aux longueurs multiples.


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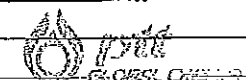
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Zugversuch Streifenprobe / Tensile test Strip test specimen / Essai de traction Bande decoupees sur tube

Pos. Item Poste	Proben-Nr. Specimen No. No. de l'éprouvette	Schmelzen-Nr. Heat No. No. de coulée	Probenabmessung Specimen dimensions Dimensions de l'éprouv.	Streckgrenze Yield strength Limite élastique	Zugfestigkeit Tensile strength Résistance à la traction	Dehnung Elongation Allongement	Einschnürung Area reduction Coefficient de striction	Re/Rm Re/Rm Re/Rm
Anforderungen Requirements Exigences			mm	ReH N/mm ² MIN 240	Rm N/mm ² 415-790	A2 [*] % MIN 27	1. Formel 1. Formula 1 Formule	
0010	000001	552918	19,00 X 4,00	330	473	36		
0010	000002	552918	19,00 X 4,00	320	474	34		
0020	000001	762106	25,40 X 5,60	318	467	40		
0020	000002	762106	25,40 X 5,50	326	471	42		
0030	000001	762135	25,40 X 5,90	317	483	36		
0030	000002	762135	25,40 X 6,10	320	477	40		
0030	000003	762135	25,40 X 6,00	338	486	42		



Chayut S.

Inspection QA/QC Data:

Integrity and Reliability

Kerbschlagbiegeversuch Notched bar impact test / Essai de flexion par choc (résilience) [CHARPY_V]

Pos. Item Poste	Proben-Nr. Specimen No. No. de l'éprouv.	Schmelzen-Nr. Heat No. No. de coulée	Probenabmessung Specimen dimensions Dimensions de l'éprouvette	Probenlage Specimen position Position de l'éprouvette	Prüftemperatur Test temperature Température d'essai	Kerbschlagarbeit Absorbed energy Energie absorbée	Kerbschlagzähigkeit Impact strength Résistance au choc	Verf.-Bruchanteil Shear fracture Rupture ductile
Anforderungen Requirements Exigences			Länge Length Longueur	Breite Width Largeur	Höhe Height Hauteur	längs (L) longitudinal (L) longitudinal (L)	einzel single individuelle	mittel average moyenne
			mm	mm	mm	quer (Q) transversal (Q) transversal (Q)	GRAD °C -45	J MIN 007 MIN 009
0020	000001	762106	55	5,00	10,00	L	-45	104 112 100
0030	000001	762135	55	5,00	10,00	L	-45	86 93 89

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Kerbschlagbiegeversuch Notched bar impact test / Essai de flexion par choc (résilience) [CHARPY_V]

Pos. Item Poste	Proben-Nr. Specimen No. No. de l'éprouv.	Schmelzen-Nr. Heat No. No. de coulée	Probenabmessung Specimen dimensions Dimensions de l'éprouvette			Probenlage Specimen position Position de l'éprouvette	Prüftemperatur Test temperature Température d'essai	Kerbschlagarbeit Absorbed energy Energie absorbée		Kerbschlagzähigkeit Impact strength Résistance au choc		Verf.-Bruchanteil Shear fracture Rupture ductile
			Länge Length Longueur	Breite Width Largeur	Höhe Height Hauteur	längs (L) longitudinal (L) longitudinal (L)		einzel single individuelle	mittel average moyenne	einzel single individuelle	mittel average moyenne	
Anforderungen Requirements Exigences						quer (Q) transversal (Q) transversal (Q)	GRAD °C -45	J MIN 005	J MIN 006	J/cm²	J/cm²	
	0010 000001	552918	55	3,33	10,00	L	-45	50				
				3,33	10,00			48				
				3,33	10,00			46	48			

Wärmebehandlung / Heat treatment / Traitement thermique

Normalized: 900 °C

Vermerk / Remark / Remarque

NACE Standard: Hardness acc. to NACE MR0175/ISO 15156: HRC max 22.

Verkäufer(in) / Salesman/ woman in charge / Personne chargée : Mrs Sehrbrock,

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GLOBAL DIVISION
Chayut S.
Inspection QA/QC Division
Integrity and Reliability

Chayut S.

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STAHL/ROHR
Benteler Stahl/Rohr GmbH

Abnahmebeauftragter
Inspection representative
Contrôleur
ROLF TRITTEL / WIL

Dinslaken, 02.02.2009,

Es wird bestätigt, daß die gelieferten Erzeugnisse den techn. Lieferbedingungen des Auftrages entsprechen. Dieses Dokument wurde mittels EDV erstellt und ist ohne Unterschrift rechtsgültig.
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